



Saving Record Labels \$11.8M in Marketing Waste via Predictive Analytics

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The Executive Summary

THE PROBLEM

Record labels invest heavily in song promotion without a reliable way to predict which tracks will succeed. In our data, the top 10 songs generated over \$5.9M in streaming revenue, while the bottom 10 earned almost nothing.

THE SOLUTION

We developed the model as an early A&R screening tool that classifies newly released tracks as Flop, Moderate, or Hit. Results are displayed in an internal dashboard with a performance tier, confidence score, and brief explanation, making the output easy to interpret and act on. High-confidence flops are deprioritized early to prevent labels from wasting marketing time or budget on low-potential tracks. Moderate tracks are escalated for human review, while predicted hits move forward for marketing consideration with human approval before major spending decisions.

THE PAYOFF

The model filters out the bottom 33.1% of the release slate before human evaluation. Applied to 1,000 releases, that removes 333 predicted failures and avoids about \$16.65M in annual marketing spend. After leakage and operating costs, the projected Net Business Value is \$14.37M.

The Business Problem

A&R marketing spend is a coin flip dressed up as strategy

Record labels have to choose which songs to back, but promotion costs time and money. The problem is that most songs do not perform well, and only a small number become successful.

That makes song selection risky. Labels often rely on gut feeling, an artist's reputation, or past success, but that does not always work. As a result, they can spend money promoting the wrong songs and miss better opportunities.

Because marketing and distribution account for most of the money, choosing the wrong song can be very costly.

No Early Warning System

A&R managers commit the full \$50K budget at release with no predictive signal. There is no filter between submission and spend.

Seniority Bias Hides the Real Signal

Teams default to artist fame as a proxy for quality — a self-fulfilling bias. Famous artists get marketing; unknown artists never break through.

False Negatives Cost the Catalog

Every genuine breakout act the label fails to invest in is a missed hit. The model's role is to protect the budget, not to find the next hit.

We Don't Predict Hits — We Filter Out Dead Weight

Isolating audio DNA from the artist fame confounder

01

1000 Releases Ingested

Spotify's annual release slate is used as the baseline, and tracks are cleaned and labeled by streaming outcome

02

Fame Neutralized

Artist popularity held constant - isolates whether the track itself has a commercial signal, not the name attached to it

03

3-Class Labels

Hit/Moderate/Flop is defined by streaming performance tiers, not by label input or marketing spend

04

Random Forest Trained

Handles class imbalance across Hit/Moderate/Flop, and captures non-linear audio feature interactions

05

GO/NO-GO Scoring

Flop flagged = marketing withheld
Moderate = routed to human A&R review
Hit = \$50K approved

Why not just use Artist Popularity?

The industry bets on big names, but our model shows that loudness and acousticness (not artist fame) are the strongest predictors of a flop. A high-profile artist releasing the wrong-sounding track is still a marketing risk, and this model catches that before any money is spent.

1 in 3 Releases is a Flop

Filtered out before a single marketing dollar is spent

Out of every 1,000 releases

 **333**

Tracks Filtered Out

Low-value releases removed before any budget allocation

 **666**

Tracks Filtered Out

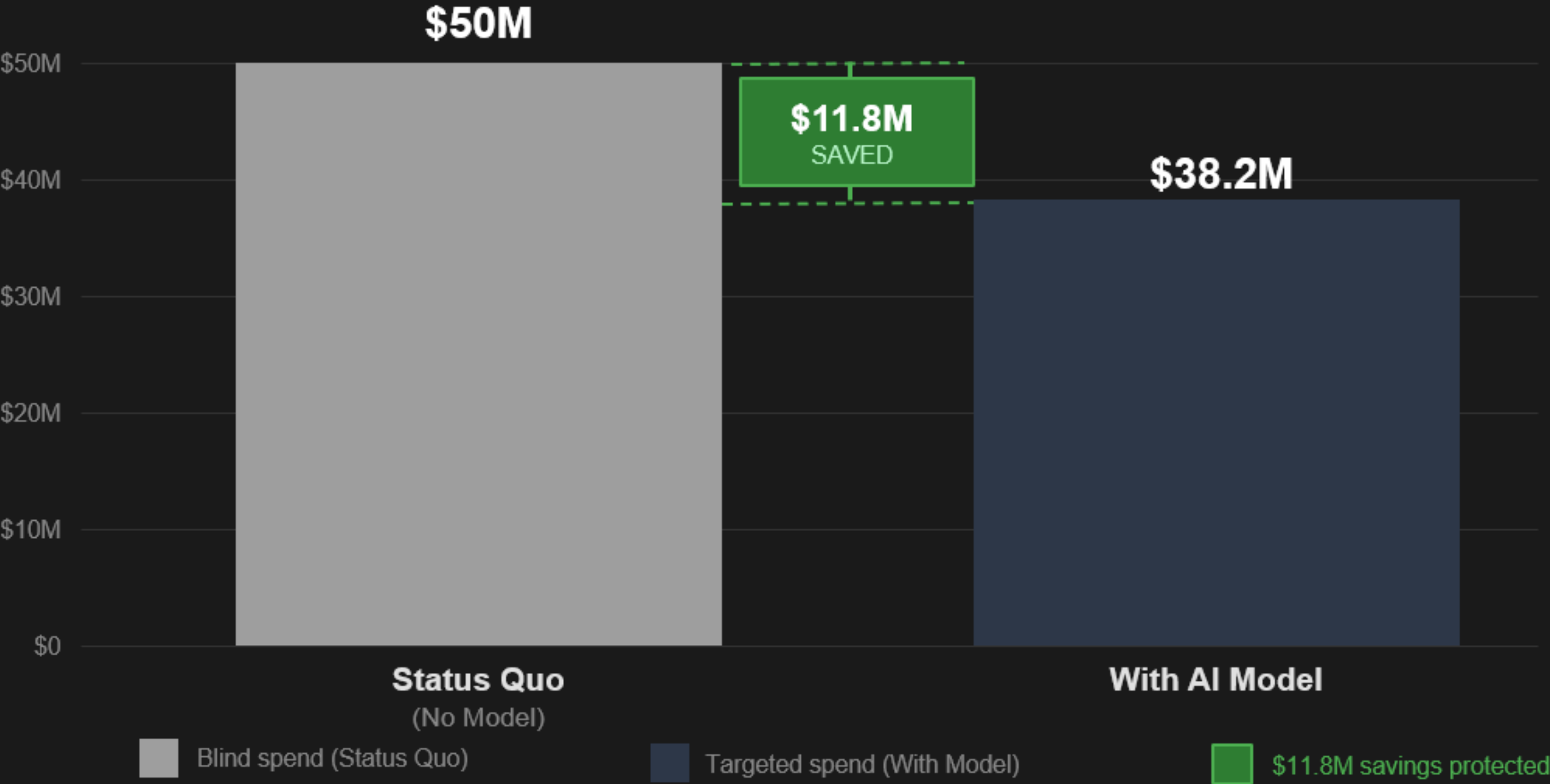
Resources and marketing budget focused where it matters

Takeaway: By filtering Flops early, we redirect analytical focus and budget to the 67% of tracks that actually drive revenue.

\$11.8M Protected Per 1,000 Releases

— From \$50M Blind Spend to \$38.2M Targeted Spend

1,000 releases · \$50K per track · Flop rate 43.5% · Flop precision 70.7% · Flop recall 54.4%



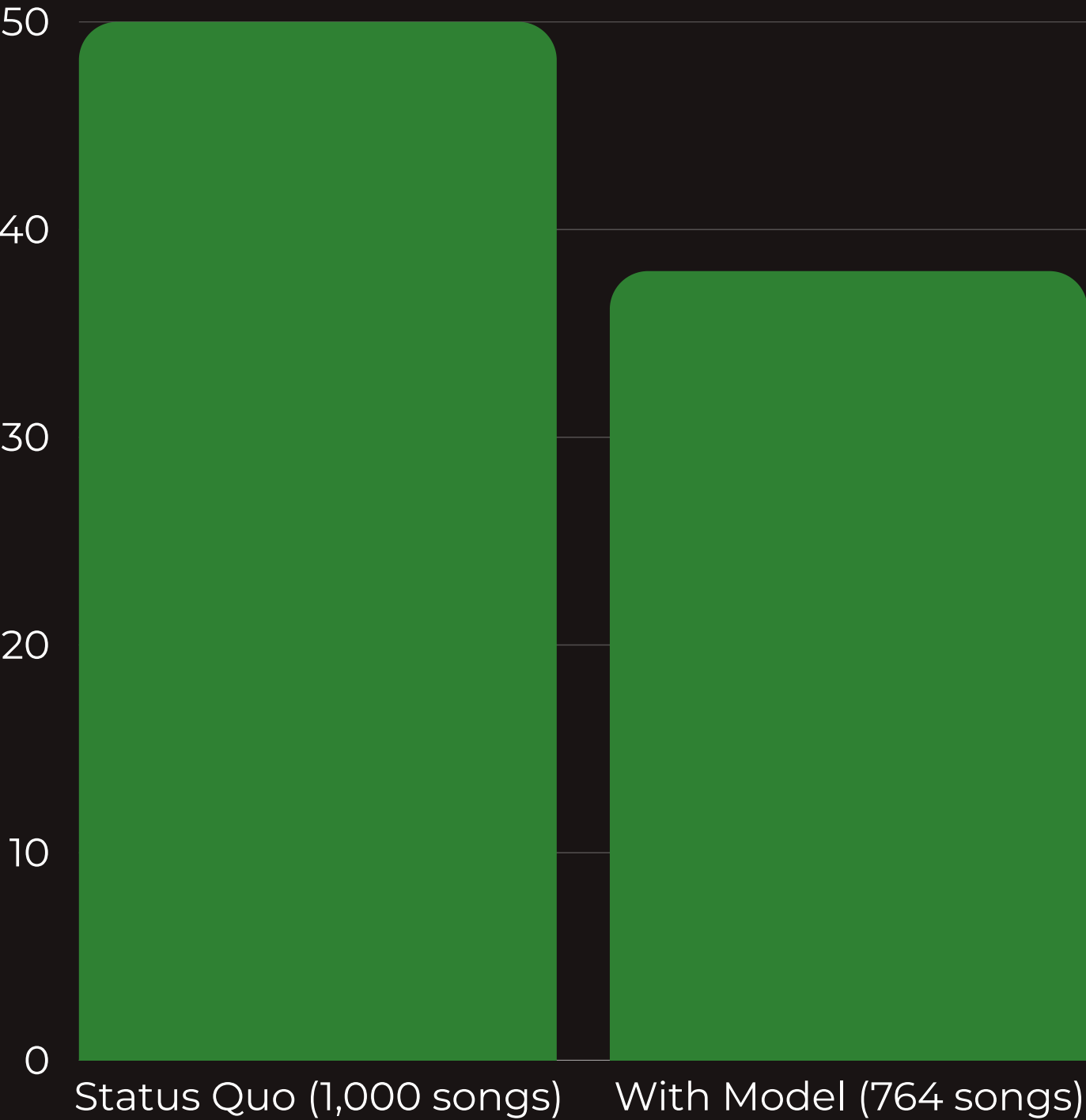
A&R Decision Analytics Dashboard



\$21.7M Lost **Annually on Low** **Potential Releases**



Total Marketing Spend (\$M)



Projected Revenue Gain & Cost Avoidance

Our predictive model identifies potential flops with **71% precision**, redirecting **\$11.8M** toward promising music releases.

Operational Risk Assessment & Mitigation Strategy

Risk 1: False Positives on Hits

\$27.7M

Annually blindly spent on low-potential tracks. Without the model, labels deploy marketing budgets before there is clear evidence of a track garnering traction.

Annual Operational Costs

\$26,200

\$1,200/yr: AWS, serverless inference, API Gateway, logging & monitoring
\$25,000/yr: Data Engineer time (20%): Model needs Quarterly Retraining & Pipeline Maintenance as song trends change.

Risk 2: Missclassified Flops

\$3.75M

The model identifies tracks with audio profiles statistically guaranteed to fail. Withdrawing spend from these releases protects the budget without requiring A&R to manually screen every submission.

Mitigation & Deployment Plan

- 1) Route every new track through the model before any marketing dollars are spent, thus withholding spend from Flops.
- 2) Reallocating \$11.8M in annual savings into higher-confidence tracks during the critical first month window.
- 3) Launch 90-day pilot: Update release checklist, train A&R on human in the loop process, & monitor precision and streaming lift.



Where the Model Breaks — and How We Contain It

Two-Stage Decision Process for Song Investment

Stage 1 - Auto - Reject ($\geq 75\%$ Flop Probability)

Songs with $\geq 75\%$ predicted flop risk are automatically removed from marketing consideration. Saves about \$50K per song in marketing costs.

Stage 2 - Performance Monitoring (Approved Songs)

Human monitoring during Month 1 performance.
~500K streams needed to continue investment.
10M+ streams = commercial success.

Where the Model Breaks — and How We Contain It

Human-in-the-loop safeguards · Oracle / Doctor / Bypass deployment modes

Model Predictions vs. Reality

- 102,788 predictions: 1,007 Hits (1.0%), 67,569 Moderate (65.7%), 34,212 Flops (33.3%).
- Actual Hits: 5,223 (5.1%) → model is very conservative in predicting success.

Key Errors & Financial Impact

- 1,007 songs predicted as Hits: 453 correct, 554 false positives.
- Equals ~\$27.7M in wasted marketing spend (\$50K per song).
- Model often confuses “hit-like” songs with real hits.

Bias & Model Limitations

- Bias: popular artists, high-energy pop songs.
- Weakness: misses new-artist breakout hits.
- Limit: cannot detect virality, marketing, or cultural timing.

DEPLOYMENT GUARDRAILS

ORACLE MODE

Flop confidence $\geq 75\%$ → auto-deferred, \$0 spent. No human required. Fast, scalable screening for clear-cut Flops.

DOCTOR MODE

Moderate & Hit-tier predictions → A&R manager review required. The model provides a signal; the human makes the final call.

BYPASS MODE

If A&R proceeds with the release, human intuition and strategic planning should guide whether to release the song as is or rework it for stronger marketing.

DRIFT MONITOR (watch daily): Avg danceability of flagged tracks. If it falls below 0.52 over a 30-day rolling window → retrain on updated release data. The shift in genre distribution is the primary leading indicator.

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DEPLOY WITH CONDITIONS:

Three Guardrails, One Clear Signal

✓ **DEPLOY WITH CONDITIONS**

PHASE 1 Weeks 1–4

Shadow Mode

Run model in parallel with current process. Flag likely Flops but don't defer spend yet. Measure false positive rate against A&R manager judgment.

PHASE 2 Weeks 5–12

Oracle-Only Deploy

Activate auto-deferral for tracks scoring $\geq 75\%$ Flop confidence only. Emerging artists remain on manual panel. Track \$ saved vs. \$ at risk weekly.

PHASE 3 Month 4+

Full Deployment

Activate all three modes: Oracle, Doctor, Bypass. Monitor danceability drift daily. Retrain model quarterly on new release data.

The model doesn't find the next Billie Eilish. It stops the label from paying \$50K to market a track that will never chart.

APPENDIX

GITHUB Repository and Code



Q&A